Connecting Computing Students' External Help Resource Preferences and Internal Help Resource Usage: 2021-2025

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Abstract

Background and Context. Academic help-seeking from both internal (course-affiliated) and external resources is a key part of computing students' learning. The recent emergence of Generative AI (GenAI) tools has substantially transformed students' help-seeking, but our understanding of this impact remains limited.
Objectives. We seek to understand the collective changes in computing students' (1) preferences of using external help resources and (2) usage of internal help resources, and the relationship between these two kinds of metrics at the individual level. We also seek to examine whether the relationship is subject to context influences.
Method. We analyzed students' self-reported preferences for external resources and recorded usage of two internal resources in 26 offerings of four computing courses ($N = 3,921$ total enrollments) at our institution over a timespan of eight 15-week terms.
Findings. We find increases in students' preferences for external resources and decreases in their usage of internal resources in some but not all courses. We identify a substantial negative relationship between the two kinds of metrics at the individual level, but its strength depends on help resource and instructional context.

CCS Concepts

• **Social and professional topics → Computing education**.

Keywords

Computing Education; Help-Seeking

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1 Introduction

Productively seeking help from resources is key to post-secondary students' successful learning in general [5] and computing [24] contexts. Considerable effort has recently been spent on studying computing students' help resource selection and preferences [4, 10, 18, 46] in the rapidly evolving landscape of help resources [19]. One way to characterize the ever-expanding set of help resources is via the *internal-external* dichotomy [18, 39]: *internal* resources are those formally affiliated with the course, e.g., class material, class-specific discussion forums [35, 37, 38, 44]) and one-on-one help from teaching assistants (TAs) [6, 15], while *external* resources (such as web-based resources [40, 47] and personal friends) are not.

Generative AI (GenAI) tools have recently transformed computing students' learning experience [30], emerging as a new form of external help resource [39]. Although students' attitudes towards using GenAI tools have yet to stabilize [29, 39], there has been an upward trend in students' usage of GenAI tools as a help resource [11, 14], accompanied with reported decreases in their help-seeking from other resources [9, 10]. These findings all support the narrative of *"GenAI is replacing everything everywhere"*.

However, our collective understanding of this impact remains limited due to (1) the difficulty of collecting behavioral data from external resources [20, 39]; (2) the reliance on self-reported usage and lack of triangulation with *individual-level* behavioral data from *internal* resources; and (3) course-specific policies [1] heavily impacting students' attitudes [39] and appropriation [29] of using GenAI tools. It is also unclear whether this impact is uniform across internal vs. external resources, different computing contexts (such as introductory vs. advanced computing courses [39]), or students of different characteristics [20]. Inspired by these gaps, we ask:

(RQ1) What is the *collective* change over time in computing students': **(RQ1a)** preferences of using *external* help resources and **(RQ1b)** usage of *internal* help resources?
(RQ2) What is the relationship between *individual* computing students' preferences of using *external* help resources and their usage of *internal* help resources?
(RQ3) Is any relationship in RQ2 sensitive to any other factors?

To tackle these questions, we study the changes in students' help-seeking approaches and behavior in four computing courses at our instituion over a timespan of four academic years (from Fall 2021 to Spring 2025, 26 offerings in total). We juxtapose students' self-reported preferences of using external resources, including but not limited to GenAI, with their recorded usage of internal resources (one-on-one help from TA and class discussion forums).

While we observe some changes in students' help-seeking over time, our results show that *GenAI has not replaced everything everywhere, yet*. Not every course had both an increase in students' preferences for external resources and a decrease in usage of the two internal resources. Even when both phenomena were observed, their timings often did not align, nor did one always precede the other. Zooming in, we identify substantial *negative* relationships between individual students' preference of using *all kinds of external resources* and usage of *both kinds of internal resources*, with the effects on GenAI being the *weakest* rather than the strongest.

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Table 1: Overview of class-level datasets and course-level demographic distributions.

Course	Type	Student body	# Consenting Students									Gender			Race					Latinx/e/a/o		Year			
			Fa21	Sp22	Fa22	Sp23	Fa23	Sp24	Fa24	Sp25	Total	M	NB	W	A	B	W	2+	O	Y	N	1	2	3	4+
Python	Service/Entry	Non-majors	177	152	183	**163**	**192**	**77**	**159**	**108**	1,211	327	10	548	271	71	431	74	9	125	757	610	258	109	86
DSA	Required	CS majors				**238**	**272**	**177**	**131**	**65**	883	475	10	376	390	55	326	48	*	88	769	594	208	49	21
Data	Elective	Interdiscip.	144	145	152	**160**	**66**	**115**	**62**	**90**	934	499	9	409	411	57	355	65	11	96	820	50	324	313	234
Algo	Required	CS majors				**202**	**123**	**220**	**93**	**255**	893	516	9	299	370	47	326	40	50	97	763	5	178	422	272

An empty cell implies the class did not collect data. Bolded classes collected students' preferences of external help resources (RQ2). Demographical distributions are aggregated by course. Gender is [M]an, Non-binary [NB], [W]oman. Race is [A]sian, [B]lack, [W]hite, Multiracial [2+], and [O]ther. Latinx/e/a/o is [Y]es and [N]o. The 4+ option for Year includes students in their fifth year as well as non-undergraduate students. Not all students responded to every demographical question. Numbers lower than 5 are replaced with asterisks.

2 Related Works

Help Resource Selection. Selecting one or more resources from available ones is a key phase in Karabenick's model of students' academic help-seeking process [13] and its extensions [26]. Although this process has been studied extensively in both general [7, 34] and computing contexts [4, 10, 32, 46], the frequent emergence of new help resources has necessitated frequent re-examination [10, 14] and inspired studies on student perceptions of help resources [19, 49] without using a set list of resources [7].

Internal Resources. Help resources affiliated with the course are called *internal* [20, 39] or *formal* [5, 18, 19]. Thanks to easily collectible data, two such resources that have been studied extensively in computing education are synchronous consulting/tutoring/office hours [6, 15, 22, 33, 41, 45] and asynchronous class discussion forums [35, 37, 38, 42, 44]. Students' utilization metrics such as proportion of users [15, 38], frequency of usage [6, 15–17, 20, 42], and type of help sought in these resources [15, 33, 44, 45], has been extensively studied. Just in the past five years, many works have studied the rich relationships between students' internal help-seeking versus their gender [20, 35, 42], race [20, 35], prior (programming) experience [17, 37, 38], major and year/standing [20], as well as the modality of help [6], using behavioral data. However, most of these works focus exclusively on one or both internal resources without also examining students' help-seeking in external resources.

External Resources. Resources not affiliated with the course, such as personal friends/family, searching static web resources [32, 40, 47], and GenAI-based tools [29, 39], are called *external* [20, 39] or *informal* [5, 18, 19]. The field's collective understanding of student help-seeking in external resources pales in comparison to that of internal resources. Due to the difficulty in collecting behavioral data [20], most related works on external resources rely on self-reported usage [9, 10, 29], attitudinal surveys [39], or observations in controlled environments [31, 40], with lower accuracy, granularity, and/or authenticity. Several works [25, 36, 48] sidestepped this limitation by integrating GenAI into the course ecosystems, which however also converted these tools into internal resources with guardrails and narrowed down their scope of use.

Neither the instructors' [23] and the students' [14, 29, 39] views on GenAI tools have stabilized, and discrepancies in instructor-imposed policies [1] may severely impact students' usage, attitudes towards usage, and self-reporting of usage. Nevertheless, recent works have shown that students' usage of GenAI-based tools is trending up [11, 14], bringing substantial impact to their preferences for other internal [10] and external [9] resources.

Synthesis/Contribution. Internal and external help-seeking are not independent parts of students' help-seeking processes [18]. Different users prefer different types of resources [20], necessitating the examination of both types. Our work attempts to triangulate students' self-reported preference for external resources with their recorded usage of internal resources and study the interplay between them over time in multiple instructional contexts.

3 Contexts and Methodology

Contexts and Participants. Table 1 summarizes our data collection at a medium-size, research-oriented private university in the US with 15-week semesters. Introduction to Python programming (Python) is both a service course for non-majors and an entry course to the CS major. Data structures and algorithms in Java (DSA) is the next course in the introductory programming sequence.[1] Algorithm design/analysis (Algo) is an advanced required course that emphasizes the analytical part of algorithm design with minimal to no programming in weekly assignments. Data science (Data) is a popular elective with a focus on applying Python data analysis packages and an interdisciplinary student body (52.7% CS majors across all offerings). Python and Data employ a flipped learning model, while DSA and Algo are lecture-based. All courses except Data have small (<30 students) lab/discussion/recitation sections faciliated by TAs. Throughout this paper, we refer to Python, DSA, Data, and Algo as *courses*, and single offerings of a course as *classes* (26 in total). Our data spans from Fall 2021 (Fa21) to Spring 2025 (Sp25) for a total of eight semesters, although not all kinds of datasets were collected for every class.

GenAI/LLMs Usage Policies. No course in our data scope completely prohibits the usage of GenAI/LLMs tools in learning.

In the two introductory programming courses (Python and DSA), students are told to treat GenAI/LLMs as a help resource to aid learning, i.e., using GenAI/LLMs is allowed in formative assessments but not in summative assessments. Students in these two courses are instructed to *use GenAI/LLMs for concept-level help only, while all submitted code should be students' own work.*

The other two more advanced (and less programming-focused) courses do not explicitly prohibit the usage of GenAI/LLMs to generate code or mathematical proofs, *but students are considered ultimately responsible for all work that they turn in and are expected to be able to explain all details.* These policies have stayed consistent in all four courses since the emergence of GenAI/LLMs in Sp23.

[1] Many students with prior programming experience skip Python and directly enter the major by taking DSA. Therefore, we do not call them "CS1"/"CS2" as they are not necessarily the first and second course taken by a student.

Help-seeking behaviorial data in internal resources. Throughout the paper, we use *consulting hours* (CH) to refer to a designated time/place (which might be online) where students could seek *one-on-one* help from a class staff member. All classes hosted regular CH multiple days a week via a queueing app, from which we obtained each student's total number of CH Requests in each class.[2]

All classes used Ed Discussion [3] as a discussion forum for students to seek asynchronous help from the teaching staff or peers. For each student in each class, we obtained the number of threads they posted (Forum Posts) throughout the semester.[3]

Student preferences for external resources. We started surveying students in all courses on their preferences for all available help resources (including internal and external) from Sp23. The survey instructed students to describe their preferred help-seeking process by grouping all resources based on the order they would use them, *assuming all resources were available*.[4] Students put one or more resources they would use first (assuming availability) in Group 1, then listed the next batch of resources they would turn to if the first group was unavailable or unhelpful in Group 2, and continued this process for up to three groups, optionally leaving resources that they did not consider useful not assigned to any group.

In Sp23, the survey had a Online Resources option that covered all online external help resources. In light of the emergence of GenAI/LLMs help resources, this option was further separated in Fa23 into Static Online Resources and GenAI/LLMs by the distinction of whether the help resource is *interactive* (i.e., can respond according to different user inputs). All classes also included a Classmates option and a People Unaffiliated with Class option.[5]

Methods. We follow the vast majority of related works in this area [16, 17, 20, 35, 37, 42, 43] by using exclusively non-parametric statistical methods on student help-seeking metrics.

Following our other works [18, 20] that analyze the preference survey, we use the (normalized) Borda Score metric to describe students' collective preferences for a help resource in a class for **RQ1a**. Each response assigns a Borda Score of 3, 2, 1, or 0 to a resource based on whether the student placed it in their *used first*, *used second*, *used third*, or *omitted (would never use)* category, respectively. The normalized Borda Score for a resource in a class is then defined as the mean Borda Score across all students in that class. This normalization allows for fair comparison between the Borda Scores of the same resource across different classes of various sizes.

We note that the normalized Borda Score is only used for aggregating students' *collective* responses. To analyze students' preferences at the individual level (RQ2), the individual responses are treated as an ordinal variable (from *used first* to *omitted*) without numerical values. As such, we used the Jonckheere-Terpstra test [12] for ordered differences such that the hypotheses preserve the ordinality in these responses. For example, one possible alternative hypothesis between (students' responses to) Classmates and CH Requests is "the *earlier* a student prefers seeking help from Classmates, the *less frequently* they use consulting hours".

Following our recent works on analyzing student help-seeking in various instructional contexts [17, 20], no statistical test in this study draws from multiple courses. For each (external resource, internal metric) tuple, we conduct a test for each *course* (combining all class offerings of the same course), followed by a Holm-Bonferroni correction across all 4 tests to control the overall type-I error rate at the 0.05 level. We choose this plan for RQ2 to allow for the discovery of potentially different (or even contradicting) effects in different instructional contexts. After obtaining all results for individual courses, we compare the results across different help resources and different instructional contexts to answer RQ3.

We report the common language effect size (CLES) [27] for all effects. The CLES for the Jonckheere-Terpstra test is the proportion of pairs (of students with different levels of the indepedent variable) in which the student with the higher response has a higher metric (if the effect is positive) and vice versa. As such, the CLES for the Jonckheere-Terpstra test is a general case of the CLES in conventional Mann-Whitney U tests [27], which is a normalization of the U statistics that is less sensitive to different population sizes.

4 Results and Discussions

4.1 RQ1: Collective changes in help-seeking

Figures 1-3 plot all aggregate metrics *by class*, with each subfigure being a course, the x-axes indicating the semesters, and the alternating shades group academic years.

4.1.1 **RQ1a***: Changes in preferences for external resources.* The Borda Score of GenAI/LLMs (Figure 1) do *not* seem to have increased between every semester. Notably, three of the four courses studied (Python, DSA, and Algo) saw a sharp increase in the Borda Score of GenAI/LLMs from Sp24 to Fa24, followed by a less substantial but still notable *decrease* from Fa24 to Sp25. In these courses, the Borda Score of GenAI/LLMs is less than or roughly equal to that of Static Online Resources in most classes, and substantially less than that of Classmates in all classes but Algo-Fa24.

On the other hand, Data, which has the most interdisciplinary student body and the more "open" GenAI/LLMs policy (see Section 3), has the most distinct help-seeking profile out of the four courses. The Borda Score of GenAI/LLMs remained rather stable in Data and consistently lower than that of Static Online Resources over the four semesters where the former is measured. In all courses, although the Borda Score of the other existing external resources (Classmates, Static Online Resources, and People Unaffiliated with Class) fluctuated, the relative ordering among all four external resources remained mostly consistent (except for a decrease in that of Static Online Resources in DSA-Sp25).

Together, the results indicate that students had not *collectively* prioritized utilizing GenAI/LLMs over seeking help from Classmates and Static Online Resources yet.

[2]Following many prior works [6, 15–17], interactions with wait time exceeded four hours or a duration outside 1-60 minutes were discarded, as they might reflect special circumstances or human errors in using the platform.

[3]We discarded all threads with a `category` unrelated to academic help-seeking, e.g., `Logistics` or `Social`. Although many prior works [16, 17, 37, 38] also analyze the number of threads each student *read* as another separate metric of engagement on class forums, this metric naturally highly depends on the overall number of threads posted to the forum. We therefore focus on students' posting behavior in this work.

[4]See our prior works [18, 20] for more discussions on this design choice.

[5]The list of internal resources differed by each class's unique internal help ecosystem and does not allow for easy comparison across offerings. As such, that part of the data is not used in this study, as we focus on connecting students' preferences for external resources and their *behavior* in internal resources instead. See our prior work [16] for complete lists of resources for many of the classes.

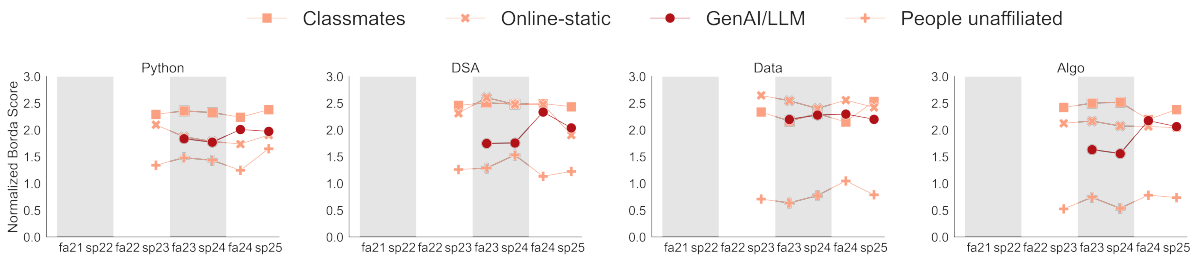

Figure 1: Normalized Borda Score of external resources by class over time. The alternating shades indicate academic years. The score for Static Online Resources in Sp23 classes is for the Online Resources option, which technically includes GenAI/LLMs.

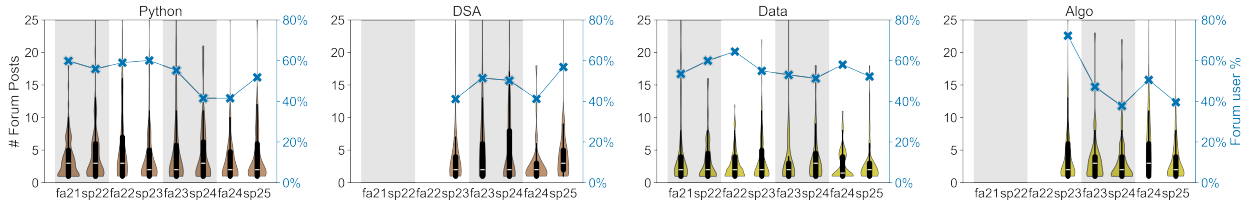

Figure 2: Forum posting rate and frequency over time. The blue line is the proportion of forum *writers* (students with at least 1 Forum Posts) in each class. For each class, the violin shows the distribution of forum *writers*' number of posts (truncated at 25).

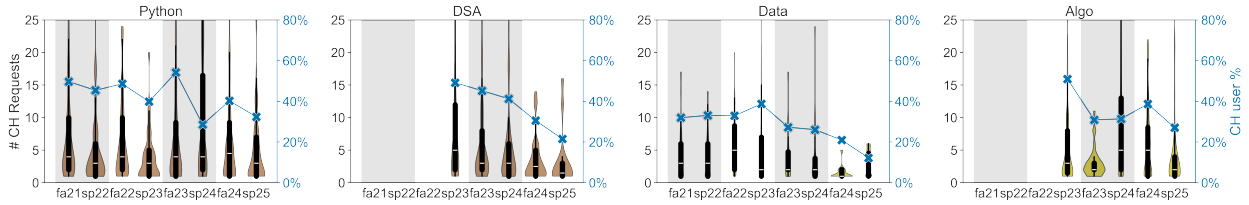

Figure 3: CH usage rate and frequency over time. The blue line is the proportion of CH *users* (students with at least 1 CH Requests) in each class. For each class, the violin shows the distribution of CH *users*' number of CH Requests (truncated at 25).

4.1.2 **RQ1b***: Changes in usage of internal resources.* For both internal resources, we characterize students' collective usage of the resource in each class by two metrics: (1) the *usage rate*, i.e., proportion of students who utilized the resource at least once out of all consenting students, and (2) the distribution of *usage frequency* of the resource among these students. As the extremely right-skewed long-tail behavior of these usage frequencies had been well-documented in prior works [6, 15, 16, 33], we determined that neither the mean nor the median alone sufficiently describes the distribution, and thereby opted to characterize the usage frequency of each resource by separating the non-users from the users.

Class Forums. We do not observe any clear decreasing trends in terms of the proportion of forum *writers* (students with at least one thread, blue lines in Figure 2). In DSA, this metric trends in the *opposite* direction as the Borda Score for GenAI/LLMs from Fa23 to Sp25, supporting that higher/more prioritized usage of GenAI/LLMs *substitutes* for seeking help from class forums. However, this phenomenon is not seen in other courses, and Algo saw the propotion of forum writers trend in the *same* direction as the Borda Score for GenAI/LLMs over the same period. Regarding the distribution of Forum Posts *among users* (violins in Figure 2), the median hovered around 2-3 threads in all classes, and only DSA-Fa24 and Data-Fa24 had a "flatter" violin (i.e., a notably less long-tailed distribution) than the other classes.

Consulting Hours. Compared to that of class forums, the proportions of CH users (blue lines in Figure 3) show a clearer decreasing trend over time, especially in DSA and Data where the metric steadily decreased between every pair of semesters in the GenAI/LLMs era (Sp23-Sp25). Although the metric fluctuated more in Python and Algo, the usage rates were generally lower post-GenAI/LLMs than pre-GenAI/LLMs in these two courses. The distribution of CH Requests showed a similar trend: the violins have mostly been flattening over time post-GenAI/LLMs, with Algo-Fa23 perhaps being the single outlier. Finally, we note that students' usage of the two internal resources did not always trend in the same direction (see Algo where they trended similarly[6] and DSA where they trended in opposite directions).

[6]If students' Borda Score for external resources and usage of internal resources all trended in the same direction in a course, a possible explanation would be that different offerings of the same courses induced different levels of help needs. However, that was not true for Algo (see, e.g., Borda Score of Classmates).

Table 2: Summary of Jonckheere-Terpstra test CLES values.

Metric	Forum Posts				CH Requests			
Course	Python	DSA	Data	Algo	Python	DSA	Data	Algo
N (all resources except GenAI/LLMs)	587	807	487	871	587	807	487	871
N (GenAI/LLMs)	432	570	332	671	432	570	332	671
Classmates	**−.62*****	**−.59*****	**−.54***	**−.54***	−.51	−.53	−.51	**−.54****
Static Online Resources	−.54	−.52	−.54	**−.54****	**−.58*****	**−.57*****	−.52	**−.54****
People Unaffiliated with Class	**−.57****	**−.53****	**−.55***	−.52	**−.55****	**−.56*****	**−.55****	**−.54****
GenAI/LLMs	−.55	−.50	−.53	+.50	−.53	**−.54***	−.52	−.53

Plus (+, shaded) and minus (−, unshaded) signs indicate the effect directions (regardless of significance), i.e., students indicating to use the row help resource earlier are in aggregate more likely to have a higher (+) or lower (−) usage metric. Holm-Bonferroni correction was applied across four courses *per resource per metric*. Effects significant at the 0.05 level post Holm-Bonferroni correction are bolded and followed with one asterisk (< 0.05), two asterisks (< 0.01), or three asterisks (< 0.001). Note that GenAI/LLMs was not included in the Sp23 surveys, resulting in different sample sizes.

4.1.3 Discussions. By simple descriptive statistics and visualization, we examined what happened at our institution and found that *both students' external and internal help-seeking indeed changed, but in a more sophisticated way than we thought.* Before examining the data, we expected to see (a) a sharp increase in Borda Score for GenAI/LLMs at some point between Sp24 and Sp25; (b) decreases in both the usage rates and the usage frequencies *in both internal resources*; and maybe (c) a decrease in Borda Score for other existing external resources (e.g., Classmates) corresponding to the increase for that of GenAI/LLMs.

However, *in no course did we see all three happen*, and where two of the expected phenomena happened, the timelines often did not match. As examples, we saw a sharp decrease in both kinds of internal help in Python between Fa23-Sp24 (phenomenon b) followed by an increase of Borda Score of GenAI/LLMs from Sp24-Fa24 (phenomenon a); nonetheless, the relative timings swapped in Algo, where the change in external help (a) happened Sp24-Fa24 and the change in internal help (b) happened Fa24-Sp25. As such, we then sought to investigate the relationship between students' preference for external resources and their usage of internal resources *at the individual student level* without the temporal lens.

4.2 RQ2: Relationships between external and internal help-seeking

4.2.1 Results. Table 2 summarizes the Jonckheere-Terpstra test results. The results show across-the-board *negative* relationships between students' preferences for external resources and their frequencies of using internal resources. Although not all effects were post-correction statistically significant, all were *negative* except for GenAI/LLMs vs. Forum Posts in Algo. In other words, the earlier a student indicates they use an external resource, the less frequently they seek help from internal resources. This phenomenon itself is not surprising at a high level: each time a student's help need is sufficiently addressed by a prioritized external resource, there is no longer a need to seek help from internal resources on the same issue. However, our results show that this phenomenon applies to all resources that we study, and is substantial even when students *without any need to seek help* are included in the samples.

Notably, this negative relationship is the *weakest* for GenAI/LLMs among all external resources. This is evident in both significance (only one of eight effects regarding GenAI/LLMs is post-correction significant at the 0.05 level, compared to at least four for others) and effect sizes (GenAI/LLMs had the lowest-magnitude CLES in four of eight columns in Table 2 and the second-lowest for the other four).[7] In other words, among all four external resources studied, GenAI/LLMs might actually be the most *complementary* with internal help resources (as opposed to *substitutable*): using GenAI/LLMs early in a student's help-seeking approach may trigger more downstream help-seeking from internal (and usually more authoritative) resources.

4.2.2 Discussions. Recent works on students' attitudes [39] towards using GenAI/LLMs (in vignette-style, hypothetical scenarios) and appropriation of GenAI/LLMs [29] *on their own use* have yet to stabilize and are seemingly dependent on context [39] and student characteristics [29]. On the other hand, our findings reveal another part of the picture: (self-reported) prioritizing GenAI/LLMs usage does not relate to the same level of decreases in internal help usage as prioritizing other external resources do, which may imply that students, even with high GenAI/LLMs adoption/appropriation, have yet to use it as effectively as other external resources.

4.3 RQ3: Resource and context differences

4.3.1 Differences between class forums and consulting hours. Table 2 reveals a subtle difference in the RQ2 results: Classmates and Forum Posts seem to be *more substitutable* with each other (than with other resources), and same goes for Static Online Resources and CH Requests. This is evidenced by that the negative effects between Classmates and Forum Posts are stronger than that between Classmates and CH Requests, while the effects between Static Online Resources and Forum Posts are weaker than that between Static Online Resources and CH Requests.

These dynamics do not seem to be well-explained by the important *dimensions* (traits) of help resources in our framework of help resource landscape [19]. For example, Static Online Resources (high availability, low adaptability, and low socialness) and CH (low availability, high adaptability, and high socialness) are drastically different help resources in terms of these dimensions. More in-depth studies are needed to detangle the nuanced dynamics between various pairs of external and internal resources.

[7] We note that although the sample sizes for GenAI/LLMs are smaller than that for the other three external resources (due to GenAI/LLMs not captured in Sp23), this discrepancy should not biase the effect sizes for GenAI/LLMs in either direction.

4.3.2 Differences across instructional contexts. The effects in Table 2 seem to be slightly more pronounced in introductory programming courses (Python, DSA) than that in advanced, non-programming focused courses (Data, Algo). This might be explanable by the nature of the contents in these courses: it is known that GenAI/LLMs perform relatively better in simple programming tasks compared to high-level interpretation tasks (ubiquitous in Data) or open-ended mathematical reasoning tasks (Algo) [8], and thus the effective use cases of GenAI/LLMs in the more advanced courses are more likely to be limited to explanation of concepts as opposed to completing the actual work. The discrepancy in GenAI/LLMs policies in these courses (see Section 3) also reflects this context.

Beyond content differences, students in more advanced courses may also possess higher-level help-seeking skills [20] and/or more sophisticated help-seeking approaches (which would allow for using different help resources for different purposes, e.g., using GenAI/LLMs for reviewing concepts, then using internal resources for completing formative assessments). In contrast, introductory programming students, being less experienced in help-seeking and in postsecondary learning in general,[8] are more likely to lack adequate metacognitive skills to develop and maintain the same help-seeking approach: indeed, a recent survey revealed that many introductory programming instructors observed many students being over-reliant on using GenAI/LLMs for every purpose [30].

5 Limitations and Future Work

Context. Our data is from one institution with medium-to-large sized classes and a predominantly White and Asian student body (see Table 1) at a research-oriented institution in the US. The findings are therefore limited to this context and may not necessarily generalize to other kinds of contexts. Each of the four courses we studied had rather consistent and "open" policies towards using GenAI/LLMs over time (Section 3), so we lack insights in students' help-seeking under more prohibitive policies or in changing environments. However, even within our stable context, our results still revealed cross-resource (Section 4.3.1) and cross-course (Section 4.3.2) differences, further demonstrating that students' help-seeking is highly context dependent.

Data Collection. Due to fluctuations in enrollment and inconsistencies in our informed consent process, later semesters had slightly fewer consenting students than earlier semesters (Table 1), making the class-level aggregate metrics (used in RQ1) noisier in later semesters.[9] Our data on students' preferences for external help resources is exclusively self-reported and therefore subject to reporting biases and low fidelity. On the other hand, our behavioral data is limited to internal resources and still subject to potential inaccuracies (e.g., not every CH interaction was properly logged into the platform). However, there is no reason to believe that *behavioral* data inaccuracies would bias in favor of/against students with higher/lower *preferences* for external resources, as the two data sources are independent. The impact on our findings in RQ2 should therefore be negligible.

Methodology. We used aggregate descriptive statistics and quantitative methods to examine student help-seeking data. While this enables studying overall changes over time and revealing correlations between different parts of student help-seeking, our methodology is not suitable for gaining rich insights on individual students' help-seeking approach and behavior, which is shaped by their unique identity and experience in learning computing. Due to the low granularity of preference data (only four possible outcomes for each student for each resource, see Section 3), the extremely long-tailed distributions of usage frequencies, and the potential concern of context differences leading to contradictory effects, we did not attempt to unify all research questions with a longitudinal statistical model, which might have nevertheless led to more robust insights.

Scope. Student demographics such as gender [20, 42], race [20], gender/race intersectionality [35], as well as characteristics such as prior (programming) experience [17, 37] and major [4, 20] have all been shown to relate to students' help-seeking approach and behavior to various extents, but are not studied in this work. Particularly, we have observed a slight increase in students' prior computing experience levels over time in our introductory programming courses.[10] As more experienced students seek less help from internal resources [17], these increases may have also led to decreases in internal help resource usage, for an effect *independent* to the impact from new emerging external resources.

Using a broader lens of equity, how the emergence of new resources (such as GenAI/LLMs) impact specific student populations, especially minority groups with lower self-efficacy [28], sense of belonging [21], or self-assessment [2], also remains to be studied.

6 Conclusions and Implications

This work seeks to examine how computing students' preferences for external help resources and usage of internal help resources change over time, as well as how these two parts of their help-seeking are related to each other. We observed increases in students' preference for GenAI/LLMs as well as decreases in their internal help-seeking (in some but not all courses), but the timings of these did not always align. At the individual level, we observed a consistent negative relationship between students' preferences for external resources and usage frequencies of using internal resources, but the strength of this effect differs by resource and by context, and is the weakest for GenAI/LLMs among all external resources. Our results reveal nuanced interplay between different help resources and different instructional contexts, and show that the narrative of *"GenAI has replaced everything everywhere"* is not universally true. Educators interested in understanding the impact of new help resources on their students' help-seeking should take extra care in accounting for their own context.

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[8] As we report in Table 1, 57.3% of Python students and 68.1% of DSA students were first-year students, in contrast to only 5.4% and 0.5% in Data and Algo, respectively.

[9] We also note that the consent rate has not always been consistent across all demographic groups. See our recent work on the relationships between student characteristics and help-seeking [20] for discussions on how this could bias the data.

[10] For example, 71.8% of students in DSA-Sp23 indicated they had less than a full course of Java programming experience, while only 10.9% reported more than a full course. In DSA-Sp25, the former decreased to 57.5% while the latter increased to 23.7%.

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